

Improved power management control strategy for renewable energy-based DC micro-grid with energy storage integration

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Abstract: This study presents an improved power management control strategy of a hybrid direct current (DC) micro-grid (MG) system consisting of photovoltaic cell, wind turbine generator, battery energy storage (BES), fuel cell (FC), and electrolyser. Based on the voltage and state of charge of BES, FC, and electrolyser, the proposed control scheme improved the dynamics of the DC-link voltage and contributes a better power management between each generation/source and load. A gain control technique is implemented in the grid-side inverter controller to regulate the modulation index and improving the voltage stability of the DC-link. Furthermore, the PI-controller gains of BES are tuned dynamically based on the deviation in voltage and its derivative using Takagi–Sugeno-fuzzy control to enhance the transient response of the voltage. For a reliable operation of the DC MG under standalone or prolonged islanding mode of operation, a priority-based load shedding algorithm is proposed for maintaining proper power coordination between different energy sources and storage devices. Owing to smoother and faster voltage response, the proposed control schemes can comply with the grid code requirements of the changing configuration of the modern renewable energy integrated DC MG. The effectiveness of the proposed control strategy is tested by comparing the existing scheme through MATLAB/Simulink®.

Nomenclature

G	ideality factor of photovoltaic cell
q	electron charge of photovoltaic cell
k	Boltzmann constant
n_s	number of solar panels connected in series
n_p	number of solar panels connected in parallel
I_{pv}	output current of the solar panels
I_{rs}	reverse saturation current in photovoltaic cell
V_{pv}	output voltage of the solar panels
T_p	surface temperature of the solar panels
R	radius of wind turbine blade
C_p	power coefficient of wind turbine
I_{dc_opt}	optimum DC-link current of wind system
V_{dc_opt}	optimum DC-link voltage of wind system
V_w	wind speed
β	wind turbine pitch angle
ρ	air density
ω	turbine angular velocity
λ	tip speed ratio
λ_{opt}	optimum tip speed ratio of wind power
ω_{opt}	optimum rotor speed of wind turbine
n_f	number of fuel cells connected in series
T_{fc}	absolute temperature of fuel cell
E_{if}	ideal standard potential of fuel cell
n_e	number of cells of electrolyser
T_e	temperature of electrolyser
$V_{rev,e}$	reversible cell potential of electrolyser
A_e	electrode area of electrolyser
A	exponential voltage of battery
B	exponential capacity of battery
H	polarisation voltage of battery
Q	battery capacity
i_{bat}	battery current
R_{bat}	internal resistance of battery
P_{AC}	AC-load power, kW

P_b	battery power, kW
P_e	electrolyser power, kW
P_{fc}	fuel cell power, kW
P_g	combined power generated by PV and wind
P_G	utility grid power, kW
P_L	DC-load power, kW
P_{pv}	PV power, kW
P_w	wind power, kW
V_{dc}	DC-link voltage, V
V_{dc}^*	rated value of the DC-link voltage, V
V_L	line voltage on the AC-side
m	PWM modulation index of the grid-side inverter

1 Introduction

Nowadays, renewable energy sources like solar, wind, tidal, biomass, or small-scale hydro-based distributed generations (DGs) are gaining popularity as clean sources of energy [1]. DGs are limited to a few kilowatts to megawatts and are interconnected at the distribution substation, distribution feeder, or to the customer load. As the penetration of DGs increases, to manage the power, coordination between generation and demand becomes more challenging in the distribution system due to their highly intermittent nature. Therefore, the concept of micro-grid (MG) has been developed as an effective way to support a reliable and better power system operation of DGs and storage devices via distribution feeders to smaller loads. MG platforms provide several advantages such as uninterrupted service to customers, improved reliability, power quality, and operational optimality over conventional distribution systems for both grid-connected and islanded mode of operation [2, 3].

An extensive research works have been carried out on various control aspects of AC MG systems such as grid integration [4], grid-connected and islanded mode of operation [5], power-sharing among multiple energy sources [6, 7], protection [8, 9] etc. On the other hand, in the recent years, direct current (DC) MG system has attracted increasing popularity due to the requirement of single-

stage converter cost when compared with AC MG and easy controls the phase and reactive power control [2]. Apart from the above advantages, DC MGs can provide a better power quality performance during voltage sags or blackout scenarios in the utility grids [2]. A typical DC MG consists of mainly four power point terminals, namely, generation, load, energy storage devices, and grid interface [2, 3]. For a stable and economic power system operation with maintaining the DC-bus voltage constant, the optimum power flow among these terminals should be necessary during power system contingencies. Few works have been reported on hybrid energy sources-based DC MG for voltage stability and controlling the uninterruptible power management in residential complexes, telecommunication buildings, Internet data centres etc. [10–12].

In [13], a power flow management and voltage control scheme is proposed in an isolated DC MG consisting of photovoltaic (PV) and battery storage device. However, in this scheme, the grid-connected and islanded mode of operation with consideration of other renewable sources and energy storage systems are not considered. The operation and power management strategy of a DC MG consisting of only PV and battery storage system is proposed in [14]. In this scheme, the maximum renewable energy utilisation is achieved during different operating modes of the MG by considering the DC voltage control, battery power management, state of charge (SoC) of the battery, and DC load. However, other renewable sources and energy storage systems are not included in this study. A reliable power control scheme in DC distribution system for balance and line fault conditions are highlighted [1]. However, the scheme is limited to a DC distribution system consisting of a wind power generation system, battery, and DC loads.

In [15], a voltage variation-based control strategy and frequency band-based power-sharing scheme are proposed for variable wind generation and multiple slack terminals of DC MG system. In [3], the authors have proposed a three-level autonomous control based on a DC voltage variation scheme for the same DC MG system as presented in [15]. The performance of the control scheme is evaluated for different operating conditions such as load shedding, generation curtailment, and generation fluctuation. Another voltage control scheme based on fuzzy control with gain scheduling technique is proposed in [16] for accomplishing both power-sharing and energy management task in a DC distribution system. However, the control strategies for the fuel cell (FC) and dump load (i.e. electrolyser) are not considered in [3, 15, 16].

In [17], the voltage droop control strategy is proposed to achieve a stable load sharing among the load and sources in the DC MG using a quadratic and linearly constrained optimisation problem. However, the authors have taken a linear model system and not considered the voltage performance during the fault scenario of the MG in the grid integration mode of operation. In [18], the authors have proposed an optimisation technique for the renewable DG using ant-lion optimisation technique. Mainly, the authors have focused on the optimal location and size of DG in a radial distribution system. In this study, the voltage performance of the PV and wind-based MG using this optimisation technique is evaluated under variable load condition. However, the authors have not highlighted the voltage performance of the MG by considering the energy storage elements (i.e. FC, battery), dump load, and grid-connected system. In [19], the authors have proposed an energy management scheme of DC MG, which consists of PV, diesel generator, and supercapacitor-based energy storage. However, wind and energy storage systems are not considered in this study. Using a sliding mode scheme for DC bus voltage control for a standalone system consisting of PV and battery is presented in [20]. However, in this study, wind power, FC with grid-connected scenarios are not taking into consideration. The authors are presented in the literature for the hybrid AC–DC MG focusing on each of the broad aspects of control, namely modelling, power management, coordinated control, voltage stability analysis, power quality, and protection strategies [21]. In [22], the authors have focused on fast transient response, low fluctuations of DC-link voltage, good rejection of disturbance due to the sudden change in active power for the grid-connected mode. However, the energy storages, dump

load, islanding mode, as well as fault conditions are not considered in this study.

The above study clearly shows that the dynamic power management performances of the DC MGs consisting of multiple renewable sources and energy storage devices have not evaluated for grid-connected, islanded mode of operation, and line fault scenarios. For an effective power management and to cope with the increasing power demand profile, it is necessary to propose an improved power coordination strategy in renewable energy-based multi-energy storage DC MG system. In this paper, wind and solar energy-based DC MG system with three energy storages, namely battery, FC, and electrolyser are considered as shown in Fig. 1. Out of the three energy storages, the battery can act either as a source or as a sink. As a result, it can charge (discharge) within its specified limits with respect to the surplus (deficit) of power generation. The FC is only acting as a source when the availability of hydrogen is sufficient and the electrolyser acts as a sink for the system. In [23], the authors have highlighted that the voltage-based control scheme as presented in [2] has the limitation in achieving proper power coordination among battery, FC, and electrolyser.

In this paper, both voltage and SoC sensitive-based control scheme of battery, FC, and electrolyser are proposed to coordinate the power management in an effective manner. It results in improving the dynamic performance of the DC-link voltage during system contingencies. Since battery storage device acts as an important energy storage device to enhance the DC-link voltage response of the DC MG when compared with FC and electrolyser to provide the surplus power for balance between generation and demand. Hence, in this paper, the gains of the proportional plus integral (PI) controller are adaptively tuning by Takagi–Sugeno (TS)-fuzzy logic control with respect to system events to improving the DC-link voltage dynamic performance. It is well known that the TS-fuzzy-based control is more flexible due to its linguistic rule-based consequent which can produce an infinite number of gain variation characteristics and contributes a better solution for non-linear control problems [24, 25]. Further, a proportional gain control strategy of the grid-side inverter is implemented to keep the modulation index within a reasonably practical limit and contribute a proper real power management with maintaining the DC bus voltage constant. The performance of the proposed control strategy is investigated for both grid-connected and islanded mode of operation as well as a line fault condition and found to be superior to the existing methods. In [26], a load shedding algorithm is implemented on the basis of SoC of the battery. However, the authors have not considered the voltage level of DC MG while deciding the percentage of load shedding. Because the DC-link voltage is an important system element in DC micro-grid and its performance depend on the every system events such as the fluctuations in load, the variety of power generation/storage capacity and the change in the system parameters, etc. Hence, in this paper, both SoC and DC bus voltage level are taken into account while deciding the load shedding strategy for maintaining the DC-link voltage within its specified limits, for a reliable and stable power system operation.

The main contributions of this paper are summarised as below:

- To achieve an effective real power management scheme among the variable generations (wind and solar) and energy storage devices such as the battery, FC, and electrolyser, for mitigating the DC-link voltage fluctuations during system contingencies.
- TS-fuzzy-based controller is proposed to tune the PI gains of the battery system for achieving a smoother DC output voltage during different operating scenarios.
- Propose a proportional gain-based control scheme for regulating the modulation index of the DC MG inverter for improving the dynamics performance of the output voltage.
- To take an appropriate decision for load shedding, the SoC of battery and DC bus voltage-based algorithm is implemented.

The rest of the paper is organised as follows. The control strategies employed for different components of the proposed hybrid DC MG are illustrated in Section 2. In Section 3, the outlines of different control methods of the proposed control

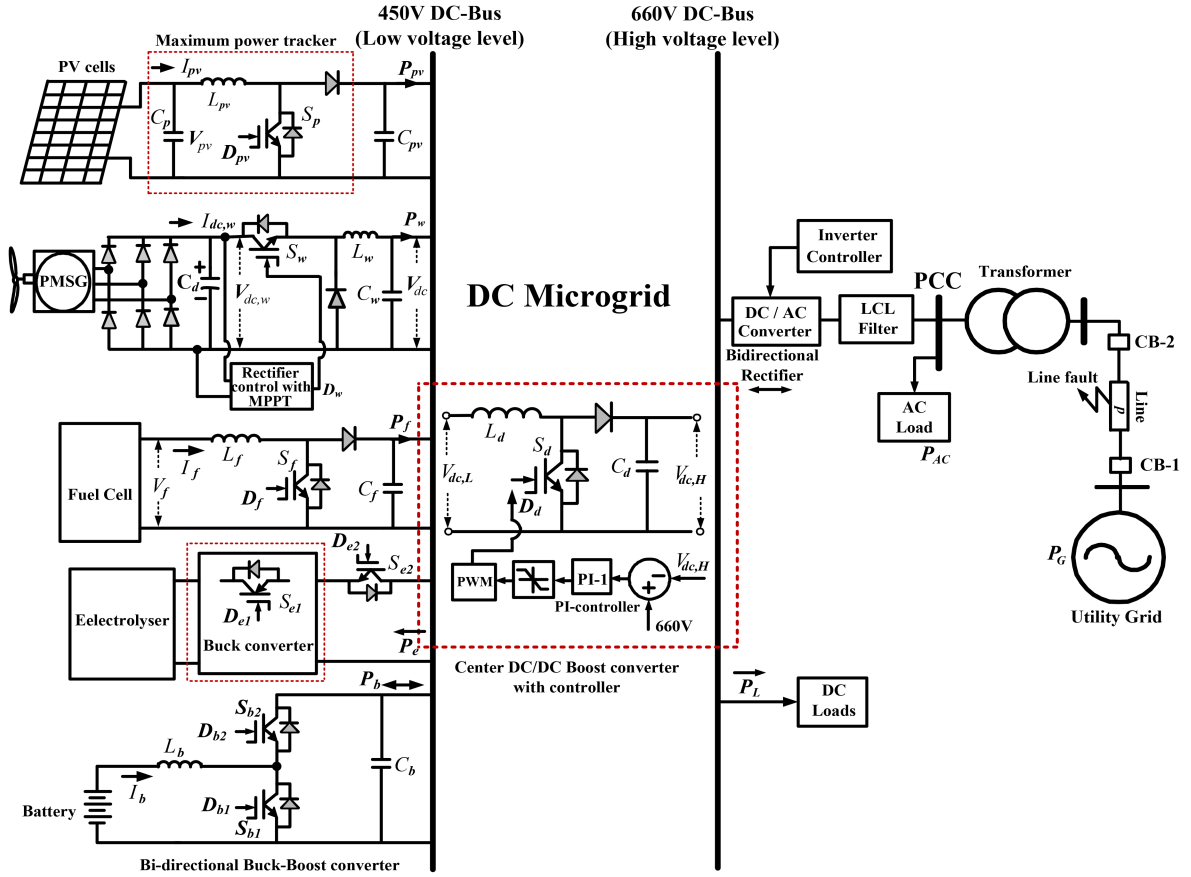


Fig. 1 Schematic diagram of the studied DC MG system

schemes are employed during different operating conditions of the hybrid DC MG. The simulation results are presented in Section 4. Finally, the conclusion of the proposed work is provided in Section 5.

2 Control strategies employed for different components of the proposed hybrid DC MG

The schematic of the hybrid DC MG system studied in the present work is shown in Fig. 1. The main components are: (i) variable renewable energy sources, i.e. PV and permanent magnet synchronous generator (PMSG)-based wind turbine system, (ii) FC and dump load, (iii) variable load scenario, (iv) battery energy storage (BES), (v) DC–DC converters, and (vi) voltage source converters (VSC). In the present study, the mathematical modelling and characteristics of each of the components used are simulated by using MATLAB/Simulink and are well documented in [14–16, 23–27]. The brief modelling of individual components of the DC MG and the proposed control strategies of the present study are discussed in this section. The detailed simulation data of the DC MG are provided in Tables 1–3. (Figs. 2 and 3)

In Fig. 1, V_{pv} and I_{pv} are the output voltage and current of the solar panels, respectively. P_{pv} is the PV power, L_{pv} and C_{pv} are the inductor and capacitor of the PV converter, respectively. S_{pv} and D_{pv} are the power electronics switch (i.e. MOSFET, insulated-gate bipolar transistor) and pulse width modulation (PWM) duty cycle of PV converter, respectively. $V_{dc,w}$ and $I_{dc,w}$ are the output voltage and current of the wind power, respectively. P_w is the wind power, L_w and C_w are the inductor and capacitor of the wind power converter, respectively. S_w and D_w are the power electronics switch and duty cycle of wind power converter, respectively. V_f and I_f are the output voltage and current of the FC, respectively. P_{fc} is the FC power, and L_f and C_f are the inductor and capacitor of the FC converter, respectively. S_f and D_f are the power electronics switch and duty cycle of FC converter, respectively. I_b is the output current of the battery, P_b is the battery power, L_b and C_b are the

inductor and capacitor of the bi-directional battery converter, respectively. S_{b1} and S_{b2} are the power electronics switch of bi-directional battery converter, D_{b1} and D_{b2} are the duty cycle of bi-directional battery converter, P_e is the electrolyser power, S_{e1} and S_{e2} are the power electronics switch of electrolyser converter, D_{e1} and D_{e2} are the duty cycle of electrolyser converter, L_d and C_d are the inductor and capacitor of the main DC/DC MG converter, respectively. S_d and D_d are the power electronics switch and duty cycle of the main DC/DC MG converter, P_L is the load power, and P_G is the grid power, $V_{dc,L}$ and $V_{dc,H}$ are low- and high-level DC-link voltage of the DC MG.

2.1 Control strategy employed for the PV cell

The output current of the equivalent circuit of the PV cell is expressed as follows [24]:

$$I_{pv} = n_p I_{ph} - n_p I_{rs} \left[\exp \left(\frac{q V_{pv}}{k T_p G n_s} \right) - 1 \right] \quad (1)$$

Similarly, the maximum output power ($P_{pv,pax}$) of the PV cell is represented as follows [15]:

$$P_{pv,max} = V_{mp} \times I_{mp} \quad (2)$$

where V_{pv} and I_{pv} are the output voltage and current of the solar panels, respectively. n_s and n_p are the number of solar panels connected in series and parallel, respectively. k is the Boltzmann constant, q is the electron charge, G is the ideality factor, T_p is the surface temperature of the solar panels, and I_{rs} is the reverse saturation current, and V_{mp} and I_{mp} are the maximum voltage and current of the solar panel, respectively.

The detailed modelling of PV cell is described in [23]. The PV system is interfaced to the DC MG through a DC–DC converter as shown in Fig. 1. The widely used perturb and observe (P&O)-based method is employed in this study, for maximum power point

Table 1 Modelling/simulation data of the DC micro-grid

DC micro-grid system/parameters	Values
<i>photovoltaic cell</i> [24]	
short-circuit current	8.21 A
open circuit voltage	33 V
maximum power voltage (V_{mp})	26.3 V
maximum current (I_{mp})	7.61 A
no. of cells per module	54
no. of cells connected in series (n_s)	10
no. of cells in parallel (n_p)	3
Boltzmann constant (k)	1.38×10^{-23} J/K
electron charge (q)	1.6×10^{-9} C
rated power	6 kW
<i>PMSG</i> [28]	
flux linkage	0.433 Vs
number of poles	10
rated speed	152.89 rad/s
armature resistance (R_s)	0.39 Ω
stator inductance (L_s)	0.0082 H
rated torque	40 N m
rated power	6.5 kW
inertia	0.01189 kg/m ²
<i>fuel cell</i> [25]	
absolute temperature (T_{fc})	1273 K
faraday's constant (F)	96,487 C/kmol
universal gas constant (R_0)	8314 J/(kmol K)
ideal standard potential (E_{if})	1.18 V
number of cells series (n_f)	215
constant (K_r)	0.842×10^{-6}
hydrogen molar constant (K_{H_2})	8.43×10^{-4}
molar constant of water (K_{H_2O})	2.81×10^{-4}
molar constant of oxygen (K_{O_2})	2.5×10^{-3}
response time of hydrogen (τ_{H_2})	26.1 s
response time of water flow (τ_{H_2O})	78.3 s
response time of oxygen flow (τ_{O_2})	2.91 sec
ohmic loss/cell (r)	$32,813 \times 10^{-8}$ Ω
rated power of fuel cell	8 kW
<i>electrolyser</i> [25]	
r_1	0.0015 Ω m ²
r_2	-6.019×10^{-6} m ² C ⁻¹
s_1	2.427 V
s_2	-0.0307 V C ⁻¹
s_3	3.9×10^{-4} V C ⁻²
t_1	0.214 A ⁻¹ m ²
t_2	-9.870 A ⁻¹ m ² C
t_3	119.1 A ⁻¹ m ² C ²
n_e	64
V_{rev}	1.1647 V
rated power of electrolyser	10 kW
<i>battery</i> [25]	
number of batteries in series	15
rated voltage of each battery	12 V
rated capacity of each battery	55 Ah

tracking (MPPT) [24]. In this study, variable nature of irradiance and temperature are considered to validate the proposed control scheme. The detailed simulation data of the PV cells are provided in Tables 1–3.

Table 1 (Continued)

DC micro-grid system/parameters	Values
<i>utility grid and distribution line</i>	
PCC voltage (line voltage)	400 V
grid voltage (line voltage)	11k V
base voltage	11 kV
frequency	50 Hz
distribution line inductance	1.05×10^{-3} H/km
distribution line resistance	0.1153 Ω /km
distribution line length	14 km
X/R ratio of distribution line	2.86
X/R ratio of utility grid	0.7
<i>transformer</i>	
rated KVA	24
rated KV	11/0.4 kV (Y–Y)
frequency	50 Hz
resistance	0.001 p.u.
inductance	0.03 p.u.
magnetisation resistance	500 p.u.
magnetisation reactance	500 p.u.
<i>DC voltage level of different subsystems</i>	
solar voltage	263 V
fuel cell	203 V
electrolyser	86 V
line voltage on AC-side	400 V
low level DC-link voltage	450 V
high-level DC-link voltage	660 V
<i>LCL filter (AC-side)</i>	
filter inductance: inverter-side (L_i)	11×10^{-3} H
filter inductances: grid-side (L_g)	0.8×10^{-3} H
filter capacitance (C_f)	12×10^{-6} F
damping resistance (R_f)	1.14 Ω
switching frequency of PWM (f_{sw})	4000 Hz

Table 2 Value L and C used in DC–DC converters (Fig. 1)

DC/DC converter	Inductance in Henry, mH	Capacitance in Farad, μ F
boost converter (PV)	5.5 (L_{pv})	2500 (C_{pv})
buck converter (wind)	10 (L_w)	1500 (C_w)
boost converter (fuel cell)	0.45 (L_f)	7500 (C_f)
buck–boost converter (battery)	12 (L_b)	1000 (C_b)
buck converter (electrolyser)	20 (L_e)	500 (C_e)
centre boost converter	0.9 (L_d)	6000 (C_d)

Table 3 Value of the PI-controller gains of the micro-grid

Subsystems	Proportional gain (K_p)	Integral gain (K_i)
boost converter		
(Fig. 1): PI-1	0.008	0.125
fuel cell		
(Fig. 2a) PI-2	0.008	0.125
PI-3	0.13	0.5
PI-4	0.0065	0.009
electrolyser		
(Fig. 2b) PI-5	80	550
PI-6	6	81
inverter controller		
(Fig. 3) PI-7	7	90

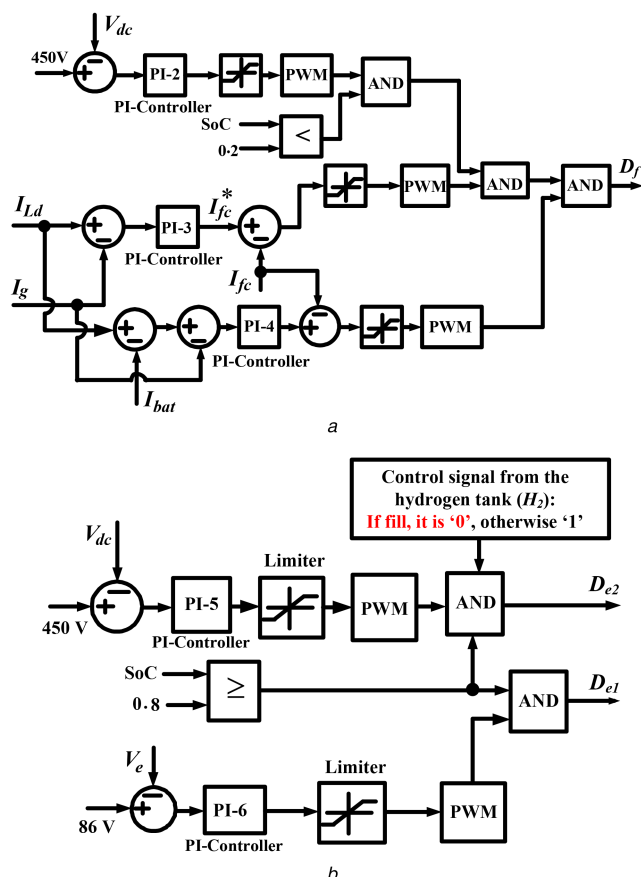


Fig. 2 Control block diagram for coordinated operation
(a) FC, (b) Electrolyser

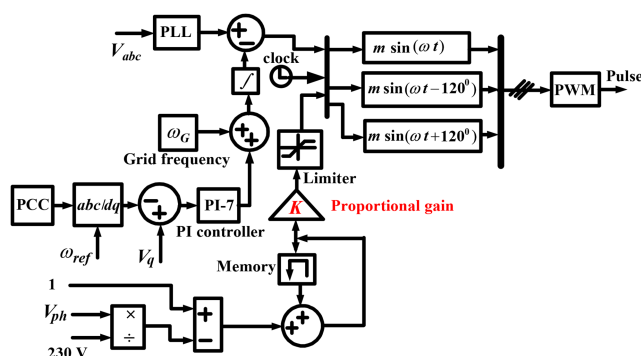


Fig. 3 Proposed control scheme during grid-connected islanding mode of operation

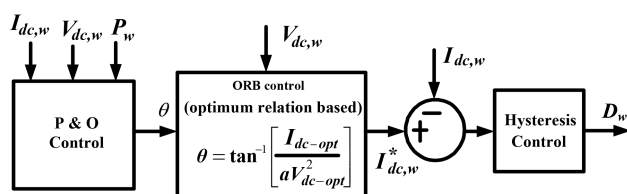


Fig. 4 Control block diagram of converter for the PMSG-based wind turbine system [28]

2.2 Control strategy employed for the wind turbine system

In this study, the PMSG-based wind turbine model used is described in detailed in [25, 27–30]. The aerodynamic mechanical power (P_w) and the maximum output power ($P_{\max}$) derived from wind by a wind turbine is evaluated as follows:

$$P_w = 0.5\rho\pi R^2 C_p(\lambda, \beta) V_w^3 \text{ and } P_{\max} = K_{\text{opt}} \omega_{\text{opt}}^3 \quad (3)$$

with

$$K_{\text{opt}} = \left[\frac{0.5 \rho \pi R^5 C_{P,\text{opt}}}{\lambda_{\text{opt}}^3} \right] \text{ and } \lambda_{\text{opt}} = \frac{R \omega_{\text{opt}}}{V_w} \quad (4)$$

where ρ is the air density, V_w the wind speed, R the blade radius, $C_p(\lambda, \beta)$ the power coefficient, which is a non-linear function of the tip speed ratio (λ) and turbine pitch angle β , λ_{opt} the optimum tip speed ratio at MPPT, ω the turbine angular velocity, ω_{opt} the optimum rotor speed at MPPT, and K_{opt} a constant.

The PMSG-based wind turbine is interfaced to the DC MG through an AC–DC stage consisting of a three-phase diode rectifier and a buck converter. Owing to the high cost and low reliability of the controlled rectifier, a simple diode-based rectifier is used in this study, for AC to DC conversion [28]. In this paper, the P&O-based control method is employed for extracting maximum power from the wind turbine [28].

The control block diagram of the PMSG set-up is shown in Fig. 4. Here, θ is used as a control parameter for perturbing. The mathematical expression of θ is defined as follows [28]:

$$\theta = \tan^{-1} \left[\frac{I_{dc-opt}}{aV_{dc-opt}^2} \right] \text{ and } a = \frac{I_{dc, rated}}{V_{dc, rated}^2} \quad (5)$$

where a is a constant and is used to scale the values of $V_{dc, \text{rated}}$ (rated DC voltage) and $V_{dc, \text{rated}}$ (rated DC current) of the wind system. $V_{dc - \text{opt}}$ and $I_{dc - \text{opt}}$ are the optimum DC voltage and current of the wind power system.

The optimum value of θ , i.e. θ_{opt} , is set close to 45° and initially, θ can be half or one-third of the θ_{opt} or less. The optimum-relation-predicated (ORB) control of wind power system for calculating the reference DC-link current ($I_{\text{dc,w}}^*$) is shown in Fig. 4 [28]. Owing to quick current controllability, fast dynamics, and less hardware requirement, a hysteresis controller is used in this study, for controlling the duty ratio of the buck converter as shown in Fig. 4 [28]. The detailed simulation data of the PMSG-based wind turbine is given in Tables 1–3.

2.3 Control strategy employed for the FC

In this study, a solid oxide fuel cell (SOFC) model is used. The detailed modelling of SOFC is presented in [24, 31, 32]. The output voltage (V_{fc}) of the SOFC is expressed as follows:

$$V_{\text{fc}} = n_{\text{f}} \left(E_{\text{if}} + \frac{R_0 T_{\text{fc}}}{2F} \left(\ln \frac{p_{\text{H}_2} p_{\text{O}_2}^{0.5}}{p_{\text{H}_2\text{O}}} \right) \right) - r I_0 \quad (6)$$

where p_{H_2} , p_{O_2} , and p_{H_2O} are the partial pressures of hydrogen, oxygen, and water response, respectively. E_{if} is the ideal standard potential, R_0 is the universal constant (J/(kmol K)); r is the ohmic losses of the FC stack (Ω), F is the Faraday's constant (C/kmol), n_f is the number of fuel cells connected in series, T_{fc} is the absolute temperature (K), and I_0 is the stack current (A).

The SOFC is interfaced to the DC MG through a boost converter as shown in Fig. 1. During normal operating condition, FC operates in standby mode. At the inception of any abnormal condition, it supplies power to the DC MG within 2 s to ensure power balance and constant DC-link voltage [24]. In [24], the authors have used a PI-controller for regulating the power of the FC through the current control scheme. However, the authors have not considered the voltage level of the FC for regulating the power. In this controller, both voltage and SoC of the FC and current comparison of generation, load have been considered to improve the dynamic power response of the FC, as shown in Fig. 2a.

In Fig. 2a, I_g is the combined generated current of PV and wind, I_{Ld} is the DC load current, and I_{fc} is the FC current in the DC MG. Whenever there is an increase in load, the extra load power is provided by the FC excluding the power supplied by the generation and the battery. The extra demanded power of the load is fulfilled

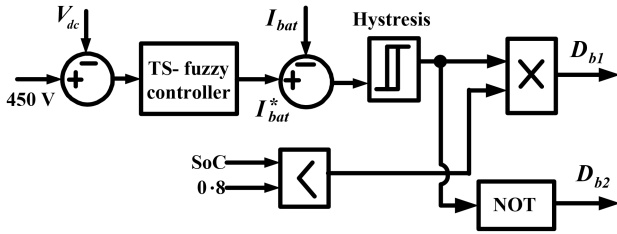


Fig. 5 Control block diagram for coordinated operation of the BES system

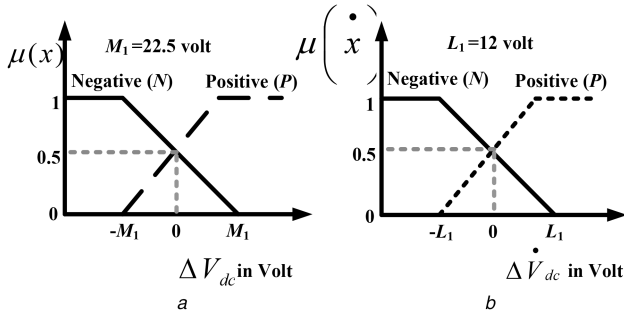


Fig. 6 Fuzzy membership functions

(a) Change in battery voltage, (b) Derivative of change in battery voltage

by the FC when the SoC of the battery is <20% and the voltage level is <450 V. The detailed simulation data of the FC are presented in Tables 1–3.

2.4 Control strategy employed for the electrolyser

In the present work, the electrolyser is used for converting the excess electrical energy generated by the wind turbine and PV cells into storable chemical energy in the form of hydrogen. The detailed modelling of the electrolyser is described in [24, 33].

The output voltage (V_e) of a temperature-dependent electrolyser model is expressed as follows [33]:

$$V_e = n_e V_{rev,e} + \frac{r_1 + r_2 T_e}{A_e} I_e + (s_1 + s_2 T_e + s_3 T_e^2) \times \log \left(1 + \left(t_1 + \frac{t_2}{T_e} + \frac{t_3}{T_e^2} \right) \cdot \left(\frac{I_e}{A_e} \right) \right) \quad (7)$$

where n_e is the number of cells of the electrolyser, $V_{rev,e}$ the reversible cell potential, T_e the temperature of electrolyser, I_e the output current of the electrolyser, A_e the electrode area, r_1 , r_2 are ohmic resistance parameters of the electrolyser, and s_1 , s_2 , s_3 , t_1 , t_2 , and t_3 the overvoltage coefficients.

The electrolyser is connected to the DC MG through a DC–DC buck converter as shown in Fig. 1. In the present work, the buck converter is regulated to feed the excess power to the electrolyser which helps in maintaining the DC bus voltage constant. The control circuit of the buck converters of the electrolyser is shown in Fig. 2b. Here, two buck converters are used as shown in Fig. 1. The function of the buck converter (S_{e1}) to maintain the electrolyser voltage 86 V and buck converter (S_{e2}) maintains the DC-link voltage (i.e. 450 V). According to the buck converter controller (S_{e2}), once the battery is fully charged (i.e. >80%), the excess power is taken by the electrolyser. When SoC becomes >80%, the controller will increase the duty cycle as a function of overvoltage. However, whenever the power demand is very less than the generation, the electrolyser will receive the excess generated power for maintaining the power balance in the DC MG.

In practice, the production of hydrogen is depend on the current of the electrolyser but not in voltage [25]. In order to stop the electrolyser to receive the excess power when the hydrogen tank is full, a control criterion is implemented which is shown in Fig. 2b. As the control signal is zero (i.e. hydrogen tank of the electrolyser is filled), the PWM pulse (D_{e2}) of the buck converter (S_{e2}) is zero,

as a result, the electrolyser disconnects from the DC MG which is shown in Figs. 1 and 2b.

2.5 Control strategy employed for the BES system

In this work, a generic Lithium-ion battery model is taken [24, 25]. The model is implemented by using a controlled voltage source in series with a constant resistance. The controlled voltage source (E_b) and the battery voltage (V_{bat}) are represented as (8) and (9), respectively

$$E_b = E_0 - H \frac{Q}{Q - \int idt} + A \exp \left(-B \int idt \right) \quad (8)$$

$$V_{bat} = E - i_{bat} R_{bat} \quad (9)$$

where E_0 is the initial voltage of the battery, H the polarisation voltage, Q the battery capacity, A the exponential voltage of battery, B the exponential capacity, R_{bat} the internal resistance of battery, i_{bat} the battery current, and $\int idt$ the charge drawn/supplied by the battery.

Further, the SoC of the battery can be estimated as follows:

$$SoC = 100 \left(1 - \frac{\int_0^t idt}{Q} \right) \quad (10)$$

The BES operates either in standby or in charging/discharging mode in normal operating condition of the DC MG. In SoC-based control method, battery charges or discharges within a specified limit whenever there is a surplus or deficit of solar energy and/or wind energy due to high or low solar irradiance/temperature and wind speed of wind turbine, respectively. The excess energy produced by the solar and the wind system is first received by the battery until it reaches its maximum limit of charge carrying capacity and then any further available of excess power is fed to the electrolyser. The switching action is carried out by comparing the present state of the battery and SoC of the battery. The control diagram of the BESS is shown in Figs. 1 and 5.

The variable generation of different energy sources and the unpredictable demand on power system impose difficulty in controlling the DC-link voltage. In the present work, the TS-fuzzy controller is used for controlling the battery voltage due to its various advantages such as flexibility in controller design, requires less processing time, and more robust for noisy input environment [34]. Using a fuzzy-based control technique, the reference battery current is estimated from the battery voltage signal as shown in Fig. 5.

In this paper, the deviation in battery voltage (ΔV_{dc}) from its nominal value, its nominal or rated value (450 V), and its deviation ($\dot{\Delta V}_{dc}$) are taken as input and actual current (I_{bat}^*) is taken as output. As a result, the value of the battery reference current (I_{bat}^*) changes adaptively with respect to the variation in voltage to enhance the dynamic response of BES voltage. In [2], it is presented that for a stable and secure operation of the DC MG, the allowable voltage excursions are about $\pm 5\%$ (i.e. -22.5 to $+22.5$ V) during system events. Therefore, in this study, the value of ΔV_{dc} is taken as $\pm 5\%$ of rated value while designing the fuzzy controller as shown in Fig. 6.

The details about the TS-fuzzy controller are given in [25, 34]. ΔV_{dc} and $\dot{\Delta V}_{dc}$ are fuzzified by using two membership functions [i.e. negative (N) and positive (P)] as shown in Fig. 6. The membership functions of ΔV_{dc} and $\dot{\Delta V}_{dc}$ for negative and positive are represented as (11) and (12), respectively

$$\mu_N(x) = \begin{cases} 1, & x < -M_1 \\ \frac{M_1 - x}{2M_1}, & -M_1 \leq x \leq M_1 \\ 0, & x > M_1 \end{cases} \quad (11)$$

$$\mu_P(x) = \begin{cases} 0, & x < -M_1 \\ \frac{x + M_1}{2M_1}, & -M_1 \leq x \leq M_1 \\ 1, & x > M_1 \end{cases}$$

$$\mu_N(\dot{x}) = \begin{cases} 1, & \dot{x} < -2L_1 \\ \frac{L_1 - \dot{x}}{2L_1}, & -2L_1 \leq \dot{x} \leq 2L_1 \\ 0, & \dot{x} > 2L_1 \end{cases} \quad (12)$$

$$\mu_P(\dot{x}) = \begin{cases} 0, & \dot{x} < -2L_1 \\ \frac{\dot{x} + 2L_1}{2L_1}, & -2L_1 \leq \dot{x} \leq 2L_1 \\ 1, & \dot{x} > 2L_1 \end{cases}$$

$$\text{with } x(n) = \Delta V_{dc}(n) = V_{dc}^*(n) - V_{dc}(n) \quad (13)$$

where $x(n)$ denotes the deviation in DC-link voltage from its nominal value at n th sampling instant, V_{dc}^* is the rated DC-link voltage, and $V_{dc}(n)$ the measured DC-link voltage at n th sampling instant.

The TS-fuzzy controller is represented by the following simplified four fuzzy rules:

Rule 1: If $x(n)$ is N and $\dot{x}(n)$ is N , then $Z_1 = b_1 \cdot x(n) + b_2 \dot{x}(n)$.

Rule 2: If $x(n)$ is N and $\dot{x}(n)$ is P , then $Z_2 = b_3 \cdot Z_1$.

Rule 3: If $x(n)$ is P and $\dot{x}(n)$ is N , then $Z_3 = b_4 \cdot Z_1$.

Rule 4: If $x(n)$ is P and $\dot{x}(n)$ is P , then $Z_4 = b_5 \cdot Z_1$.

In the above rules, Z_1, Z_2, Z_3 , and Z_4 represent the consequent of the TS-fuzzy controller, n is the n th sampling instant. b_1, b_2, b_3, b_4 , and b_5 are the fuzzy constants. The values of the TS-fuzzy constants are: $b_1 = 2.57$, $b_2 = 0.35$, $b_3 = 0.21$, $b_4 = -3.5$, and $b_5 = 3.28$.

Now, the output of the TS-fuzzy controller [i.e. reference battery current (I_{bat}^*)] is obtained by using the generalised defuzzifier, which is expressed as follows:

$$I_{bat}^* = \frac{(F_1 Z_1 + F_2 Z_2 + F_3 Z_3 + F_4 Z_4)}{(Z_1 + Z_2 + Z_3 + Z_4)} \quad (14)$$

where $F_1 = \min\{\mu_N(x), \mu_N(\dot{x})\}$, $F_2 = \min\{\mu_N(x), \mu_P(\dot{x})\}$

$F_3 = \min\{\mu_P(x), \mu_N(\dot{x})\}$, $F_4 = \min\{\mu_P(x), \mu_P(\dot{x})\}$

As the value of I_{bat}^* is regulated dynamically using fuzzy-based control which results in improving the voltage stability of power system with a better coordination of power management.

3 Control strategies employed for different power system scenarios of the proposed hybrid DC MG

The optimum energy balance between different energy sources in a DC MG is the essential requirement for maintaining the DC-link voltage within the specified range (i.e. $\pm 5\%$) during different modes of operation. The operating principles of different control strategies employed in this study to accomplish the above task are discussed below.

3.1 Control strategy employed for grid-connected/islanding mode of operation

In practice, the grid-side VSC operates for maintaining the DC-link voltage and grid-side inverter voltage within their acceptable limits

both ranges during the grid-connected and islanding mode of operation. In this work, the sinusoidal pulse width modulation (SPWM) technique is employed for controlling the VSC. The varying magnitude and phase angle of sinusoidal modulating signals are controlled by generating the modulation index (m). The relation between input DC voltage and the output AC voltage of three-phase PWM inverter is expressed as follows [25]:

$$V_L = \frac{\sqrt{3}}{2\sqrt{2}} m V_{dc} \quad (15)$$

In (15), V_L is the line voltage on the AC-side, m is the modulation index of the PWM inverter, and V_{dc} is the high-level DC-link voltage. From (15), it is seen that the AC voltage at the inverter output directly depends on modulation index. Hence, to control the modulation index, a proportional gain (K) controller is proposed, in this study as shown in Fig. 3.

In Fig. 3, V_{abc} is the three-phase voltage on the AC-side, V_{ph} is the phase voltage on the AC-side, V_q is the quadrature-axis (i.e. q -axis) voltage, ω_G is the grid frequency, ω_{ref} is the reference or rated grid frequency (50 Hz), and K is the proportional gain.

In practice, the inverter-side voltage drops or rises with respect to the power mismatch between generation and load. In the meantime, the modulation index of the inverter-side controller increases or decreases to compensate the power mismatch between generation and load. The modulation index is more oscillatory in nature when the power mismatch is more. In order to mitigate the voltage oscillation and keep the phase voltage at its rated value (i.e. 230 V), a proportional gain controller is proposed in this paper, as shown in Fig. 3. The dynamic response of inverter-side voltage and DC-link voltage of DC MG for different values of proportional gain (K) is given in Section 4.3. A memory block is used to preserve the previous value of modulation index for avoiding the voltage collapse scenario when the inverter-side phase voltage is equal to 230 V (i.e. zero-modulation index) as shown in Fig. 3. This control scheme enhances the dynamical response of voltage for both grid-connected and islanding mode of operation. To minimise the current distortion injected into the utility grid, an LCL filter is used [35]. For tracking the output frequency, PI-regulator-based phase lock loop (PLL) is used [36, 37].

3.2 Control strategy employed for load management during standalone/prolonged islanding condition

The output powers of the wind energy system and PV system vary with wind speed and temperature and irradiance, respectively. When the generated power is less than the load power, the DC-link voltage may deviate from its rated value (V_{dc}^*). If the generated power is greater than load power, the excess energy is stored in the battery and the electrolyser. When the generation power is less than the load power, the energy deficit is balanced by the FC and the battery. However, the battery capacity may be reduced below its SoC limits due to the unlikely events such as the unavailability of wind, solar power, or prolonged islanding mode of operation. In such situations, it is difficult to maintain DC-link voltage within the permissible range. In order to prevent the collapse of the DC-link voltage of DC MG, priority-based load shedding is done [26]. The flowchart of the proposed load management scheme is provided in Fig. 7 for a better understanding.

The load management algorithm of the proposed scheme (Fig. 7) works in the following steps:

- Calculate the total combined power generated (P_g) by wind power (P_w) and solar power (P_{pv}). (i.e. $P_g = P_w + P_{pv}$).
- Calculate the total DC load power of the DC MG (P_L).
- Calculate the total power difference (P_d) between generations and load ($P_d = P_g - P_L$).
- If $P_d > 0$ and $V_{dc} > 0.97 V_{dc}^*$, then check the status of SoC of the battery.
- If $\text{SoC} > 0.8$, no need of load shedding otherwise load shedding is required.

- If $P_d < 0$ and $V_{dc} > 0.97 V_{dc}^*$, then check the status of battery as well as FC.
- If $P_g + P_{fc} + P_b > P_L$, then no need of load curtailment otherwise load shedding is required based on priority of loads. Here, P_{fc} is the fuel power and P_b is the battery power.

4 Results and discussions

The performance of the proposed control method is evaluated for the different power system contingencies such as the change in wind speed, change in irradiance and temperature of the PV cell, grid-connected, islanding and line fault scenario, etc. The DC MG under simulation study comprises the PMSG-based wind turbine, PV cell, FC, electrolyser, and BES as shown in Fig. 1. The control strategies of the proposed work are briefly discussed in Section 2. The simulation data of the PV cell, PMSG-based wind power, SOFC, and electrolyser are taken from [25, 38] and the same is reproduced in Tables 1–3. The value of the proportional gain (K) of the inverter control is taken as 0.1. In the present study, the base value of irradiance and temperature of PV cells are taken as 1000 W/m^2 and 298 K, respectively. The base wind speed of PMSG-based wind turbine is set as 12 m/s. The gains of all PI-controllers are tuned by frequency response analysis and integral time square error method [39]. The comparative MATLAB simulation results of the proposed voltage and SoC-based scheme over the existing voltage-based scheme [2, 3] are presented as follows:

Scheme 1: Voltage-based DC MG control and operation [2, 3] (existing).

Scheme 2: Combined voltage and SoC-based DC MG control and operation (proposed).

4.1 Performance evaluation under fixed light load condition during standalone mode of operation

For accessing the efficacy of the proposed voltage and SoC-based control, we have considered a fixed wind speed of 12 m/s for PMSG and variations in irradiance and temperature of the PV cell as shown in Figs. 8a and b, respectively.

The dynamic responses of power management and the DC-link voltage, during a fixed light load of 3 kW, are demonstrated in Fig. 8c. In Fig. 8c, P_w is the wind power, P_{pv} is the PV power, P_e is the electrolyser power, P_b is the battery power, P_{fc} is the FC power, and P_L is the DC load power. The light load signifies that the demand is always less than the generation. However, the total

power generation, including PV and wind, is always more than the demand of 3 kW. In this case, it is considered that the battery and FC are in fully charged. During this operating condition, the electrolyser which acts a dump load takes the extra amount of power which is clearly observed in Fig. 8c. Since the power generation is excess as compared load the battery as well as the FC stops supplying power to load, in the standalone mode of operation. This gives a better dynamic response of power during light load condition in the DC MG system. From Fig. 8d, it is also seen that the DC-link voltage is maintained within the permissible range of 660 V during these operating conditions.

4.2 Comparative assessment between the proposed scheme and the existing scheme

For the performance evaluation, a comparative assessment of the proposed voltage and SoC-based control strategy is carried out with the existing voltage-based control method [2, 3] under variable loading condition. In this case study, the same variations of system parameters such as irradiance and temperature of PV with respect to time are given (Figs. 8a and b). Similarly, the change in DC load with respect to time is shown in Fig. 9a. The variable wind speed of PMSG-based wind turbine is taken as follows: for 0–4 s: 12 m/s; 4–7 s: 8 m/s; 7–10 s: 12 m/s.

The dynamic responses of the power distribution among PV cell, wind turbine, FC, electrolyser, battery, and DC loads for the above-mentioned operating conditions are provided in Fig. 9. The performances of the existing voltage-based method [2, 3] and the proposed voltage and SoC-based method are shown in Figs. 9a and b, respectively. From these figures, it is observed that the load is

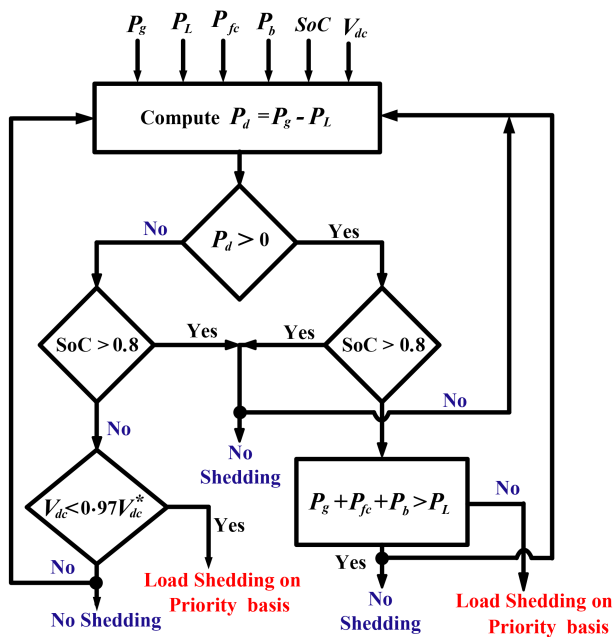


Fig. 7 Flowchart of the proposed load shedding algorithm

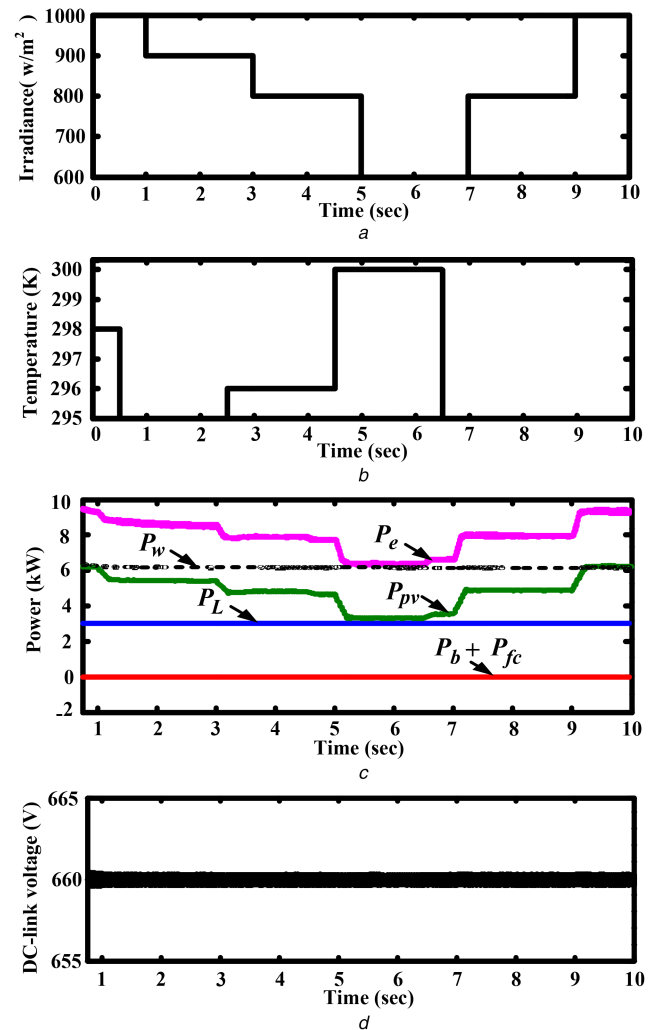


Fig. 8 Dynamic responses during standalone mode of operation (a) Change in irradiance w.r.t. time, (b) Change in temperature w.r.t. time, (c) Dynamic power responses of different sources and loads, (d) DC-link voltage response

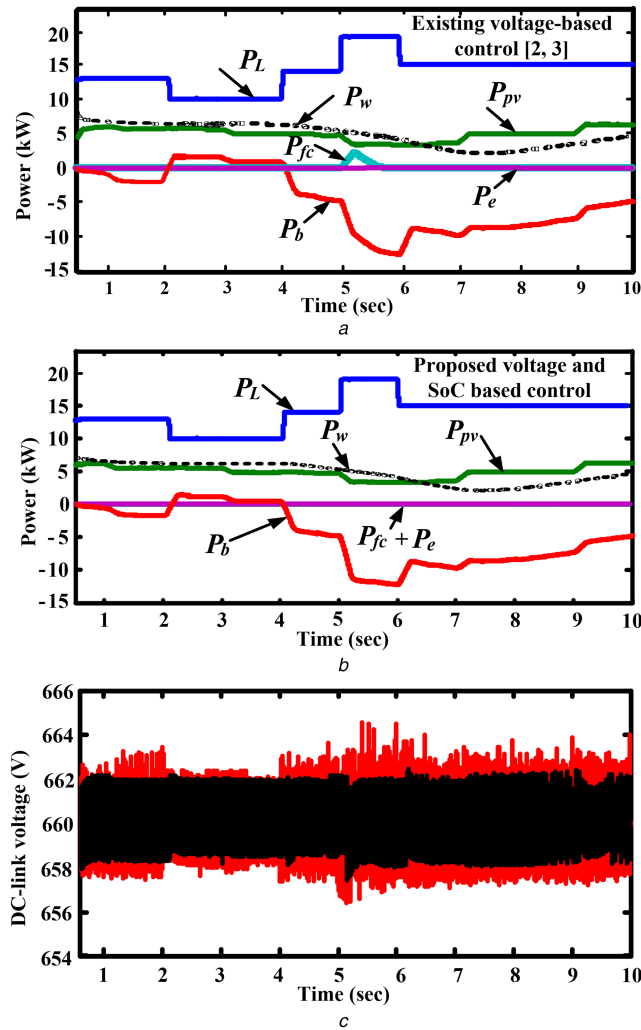


Fig. 9 Dynamic system performance

(a) Power of different sources and loads using the existing voltage control method [2, 3], (b) Power of different sources and loads using the proposed control method

Table 4 Comparison voltage deviation performance

Existing scheme [2, 3]	Proposed scheme
1.37%	0.68%

Table 5 Comparison voltage deviation performance

	Existing scheme [2, 3]	Proposed scheme
DC-link voltage	1.44%	0.91%
inverter voltage	3.51%	2.25%

fixed at 10 kW during 2–4 s. During this period, the total power generated by the PV cell and wind energy system is excess when compared with load. As a result, the battery will charge by taking the excess energy in both the control techniques (i.e. in the existing voltage-based method as well as in the proposed method). At 4 s, the load is increased to 14 kW and again, it is increased to 19 kW at 5 s as shown in Fig. 9. It can be seen that during 4–6 s, the demand of load is in excess when compared with the net power generation. In such scenario for ensuring a proper power balance between generation and load, both the battery and the FC are discharged in the existing voltage-based control method [2, 3] which is clearly observed in Fig. 9a. This shows that there is no proper coordination between the FC and the battery in the voltage-based method in Fig. 9a. For a better-coordinated operation of power distribution, the FC should remain silent (i.e. does not deliver any power) when the battery is supplying power to meet the power demand. It concludes that FC and electrolyser could not work for a particular value of voltage level. On the other hand, in

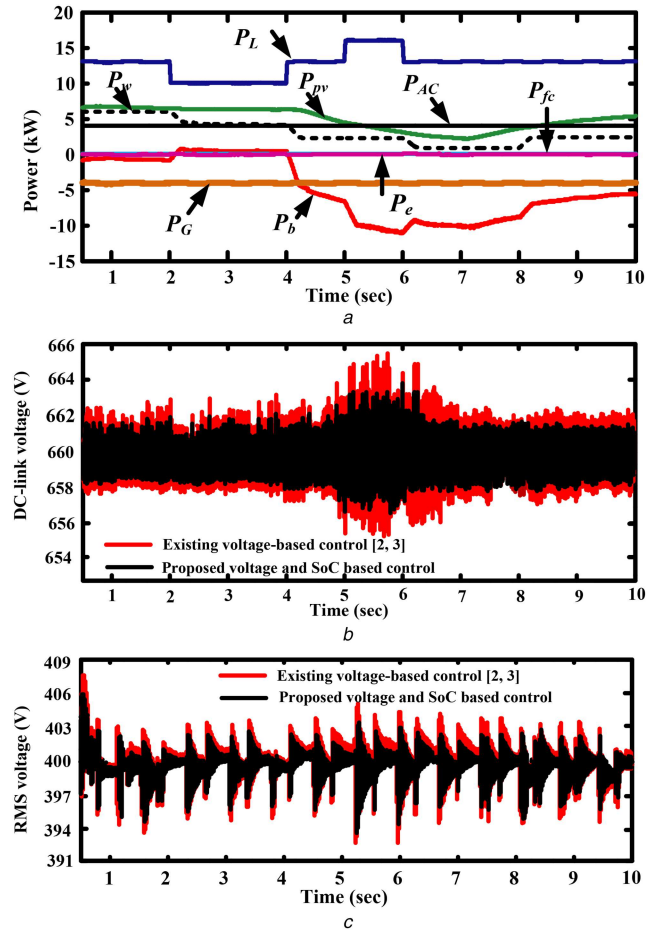


Fig. 10 Performance during grid-connected mode of operation

(a) Power of different sources, (b) DC-link voltage, (c) Inverter-side line voltage

the proposed voltage and SoC-based method, only the battery is discharged from 5 to 5.6 s to mitigate the power deficiency. As the battery is supplying power, during this period, both the electrolyser and the FC remain silent which is clearly observed in Fig. 9b. From the above discussion, it is concluded that by using the proposed control scheme, a proper power coordination between the FC, electrolyser, and the battery can be achieved.

Similarly, the comparative DC-link voltage response of the existing scheme [2, 3] and the proposed is shown in Fig. 9c. From Fig. 9c, it can be observed that the dynamic DC-link voltage response is smoother (i.e. dips/rises values are less) in the case of the proposed control scheme (i.e. black colour) when compared with the existing scheme [2, 3]. Moreover, from Fig. 9c, the exact values of the percentage of voltage deviations are mentioned in Table 4. From these data, it can be concluded that the proposed control scheme improves the overall voltage performance of the DC MG.

4.3 Performance evaluation during grid-connected mode of operation

The objective of the present study is to maintain proper power management, DC-link voltage, and inverter-side line voltage of DC MG during the grid-connected mode of operation by using the proposed control scheme as described in Fig. 3. The power of different sources and loads during the grid-connected mode of the proposed control scheme is shown in Fig. 10a. In Fig. 10a, P_G is the utility grid power, and P_{AC} is the AC-load power. From Fig. 10a, it is clearly observed that the DC-side load is fixed at 14 kW and the AC-side load is 4 kW during 4–5 s.

During this period, the total generation by the PV and wind energy system is less than the demand. As a result, the battery is discharged by giving the excess energy to maintain the power balance in DC MG and at the same time, the utility grid is supplying the required power to the local load. At this operating

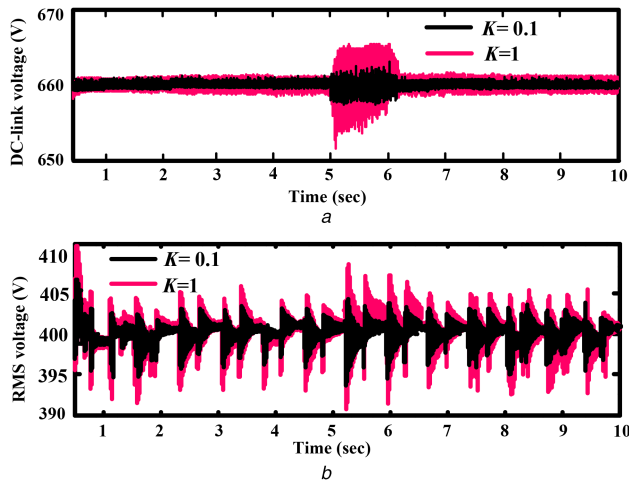


Fig. 11 Effects of proportional gain (K) on
(a) DC-link voltage, (b) Inverter-side line voltage

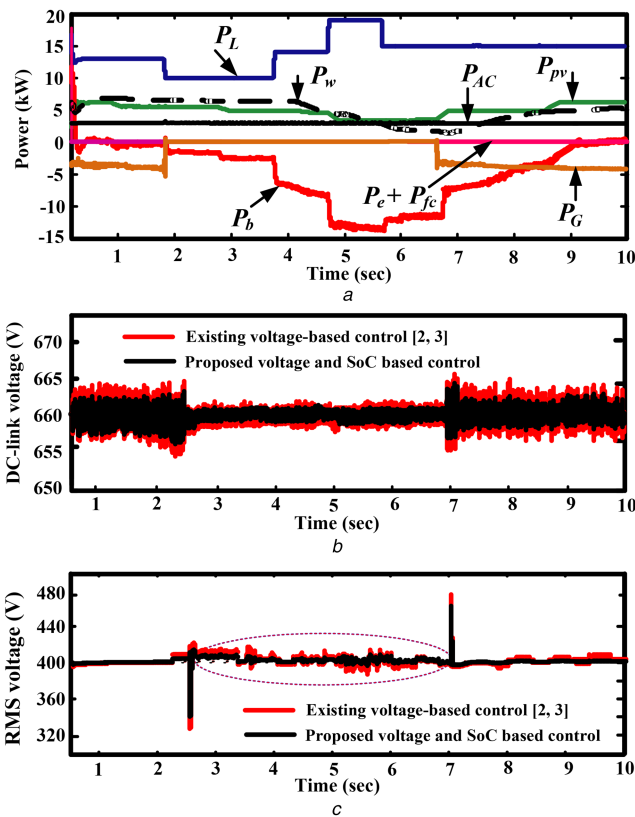


Fig. 12 Performance during islanding mode of operation
(a) Power of different sources, (b) DC-link voltage, (c) Inverter-side line voltage

condition, the DC-link voltage of the DC MG (Fig. 10b) and the inverter-side root mean square (RMS) line voltage (Fig. 10c) are maintained within their acceptable limits as shown by the black colour. Moreover, Figs. 10b and c depict that the dips/peaks in DC-link voltage and inverter-side RMS line voltage deviations are less in the case of the proposed control scheme when compared with the existing scheme [2, 3]. The exact values of the percentage of voltage deviations are mentioned in Table 5.

In order to show the effect of the proportional gain (K) on the dynamic voltage, performance of the MG is plotted in Fig. 11. For the above-mentioned scenario, the dynamic response of the DC-link voltage of DC MG and line-line root mean square (RMS) voltage of inverter-side for two different values of the proportional gain (K) is shown in Figs. 11a and b, respectively. From Figs. 11a and b, it can be observed that both the voltage responses are relatively smoother (i.e. less oscillation) in the case of the lower value of K when compared with a higher value of K .

Table 6 Comparative percentage of voltage deviations

	Existing scheme [2, 3]	Proposed scheme
DC-link voltage	1.51%	0.91%
inverter voltage	10%	4%

4.4 Performance evaluation during islanding mode of operation

The reliability of the proposed control scheme (Fig. 3) is again tested by creating an islanding mode of operation during 2.5–7 s. The corresponding simulation result is provided in Fig. 12. From Fig. 12a, it is clearly observed that at 2.5 s when the utility grid is disconnected, the proposed control scheme is able to maintain the power balance DC MG through the coordinated operation of the battery and FC. As soon as the grid is reconnected at 7 s, the battery starts charging and FC remains inactive. During this period, DC-link voltages of the DC MG, as well as the line voltage of the inverter side, are maintained within the permissible range as observed in Figs. 12b and c, respectively.

From Figs. 12b and c, it can be seen that the response of the DC-link voltage and inverter-side RMS line voltage deviations are smoother in the case of the proposed control scheme when compared with the existing scheme [2, 3]. The exact values of voltage deviations are mentioned in Table 6. Owing to smoothing in voltage response during system contingencies, the proposed control scheme improves the voltage stability of the DC MG. Hence, the proposed scheme approaches can comply with the grid code requirements of the changing configuration of the modern DC MG system integration with renewable sources. The same can be analysed for different power system contingencies in any DC MG systems. However, the dynamic response of the DC voltage, inverter-side voltage, and power coordination between each generation/source and load may vary with change in the configuration of the DC MG system, percentage of renewable energy penetration, power mismatch between generation and demand, setting of K , etc.

4.5 Performance evaluation during three-phase (LLL) fault in the utility grid

To validate the performance of the proposed control scheme during a fault condition, a short-circuit three-phase fault (i.e. LLLG) is created on the grid-side at 3.3 s from a distance of 14 km after the PCC which persists for 0.4 s. During the fault (i.e. CB-1 opens), the power supplied from the utility grid is reduced to zero except a small kink which is shown in Fig. 13a. However, the power supplied to the DC load remains constant. During this period, the deviation of the DC-link voltage is within the permissible range ($\pm 5\%$) of the nominal value as shown in Fig. 13b. In Fig. 13c, it can be observed that a large voltage drop occurs at the PCC during this fault scenario. As soon as the fault is cleared at 3.7 s, the line voltage of the PCC increases and reaches its nominal voltage. During the three-phase fault condition, the corresponding bi-directional inverter-side current is as shown in Fig. 13d. From the above results, it can be observed the deviations in system variables such as DC-link voltage, inverter-side current are within their acceptable limits during the three fault which improves the stability and reliability in the DC MG.

4.6 Performance evaluation for load shedding during standalone mode of operation

DC MG supplies power to the local loads during the standalone mode of operation. During such operating condition, when the power demand is more than the generation or curtailment of generation, the priority-based load shedding is carried out for maintaining the DC-link voltage within its specified limits and improves the stability of the power system.

To measure the efficacy of the proposed control algorithm during standalone mode of operation, four loads of 3150 W (i.e. 12,600 W) are taken at the 660 V DC-bus (Fig. 1). Each 3150 W load is the sum of four subloads of 650 W (priority 1), 750 W

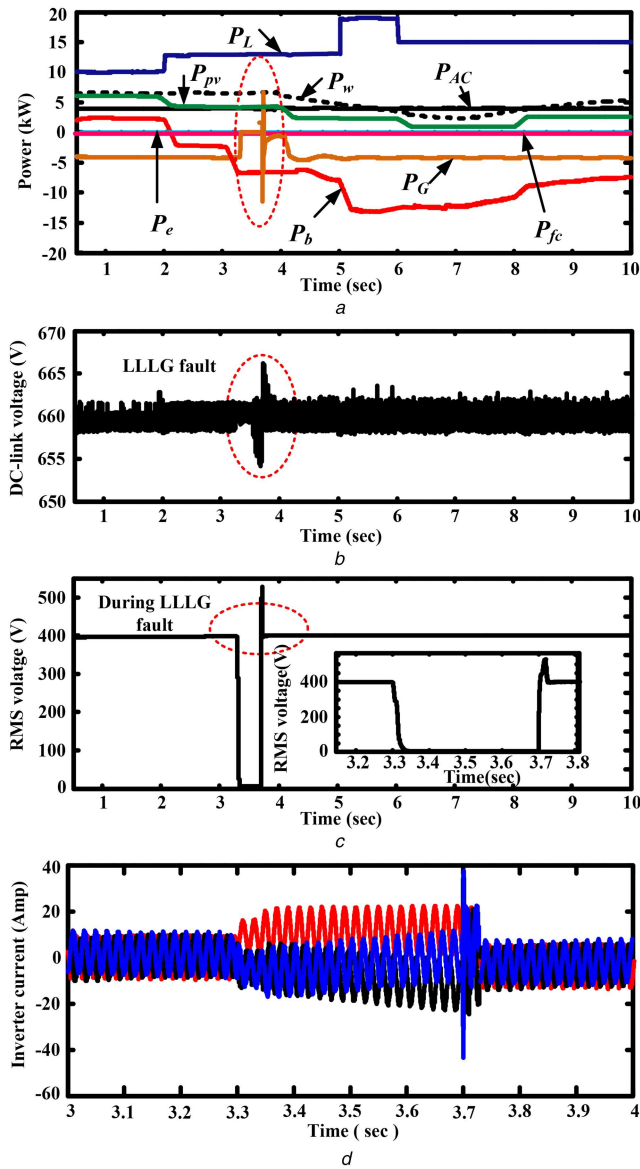


Fig. 13 Performance during the line fault
(a) Power of different sources, (b) DC-link voltage grid, (c) Inverter-side line voltage, (d) Bi-directional inverter-side current

(priority 2), 850 W (priority 3), and 900 W (priority 4) of, respectively, based on load shedding. The 3150 W load is a sum of four subloads of 650, 750, 850, and 900 W of priority 1, priority 2, priority 3, and priority 4, respectively. In the proposed method as discussed in Section 3.2, the load shedding task is accomplished by considering the difference in power level, the DC-link voltage of the DC MG, and SoC of the battery. The simulation results during the standalone mode are provided in Fig. 14.

In Fig. 14a, it is observed that the DC-link voltage is decreased more than its permissible limit of its nominal value (i.e. 660 V) in 6.1 s due to insufficient power of the battery, wind, and PV system. Hence, for maintaining the DC-link voltage of a DC MG, a load of 650 W (priority 1) of load 1 is cut off (Fig. 14b) automatically at 6.1 s as per the proposed load shedding algorithm. Again, at 6.63 s, the DC-link voltage is decreased due to further reduction in generation (Figs. 14c and d) and restricted battery power (Fig. 14e). This causes a further load shedding of 750 W (priority 2) of load 1. At 6.75 s, the DC-link voltage returns to 660 V and all the cut-off loads are reconnected. After 6.75 s onwards, there is an increase in generative power (Figs. 14c and d) and FC power (Fig. 14f) which helps in maintaining the DC-link voltage (Fig. 14a). From the above discussion, it can be found that the proposed control algorithm works in a more effectively way for sever system contingencies such as emergency demand response, prolonged islanding condition, etc. The same can be analysed for

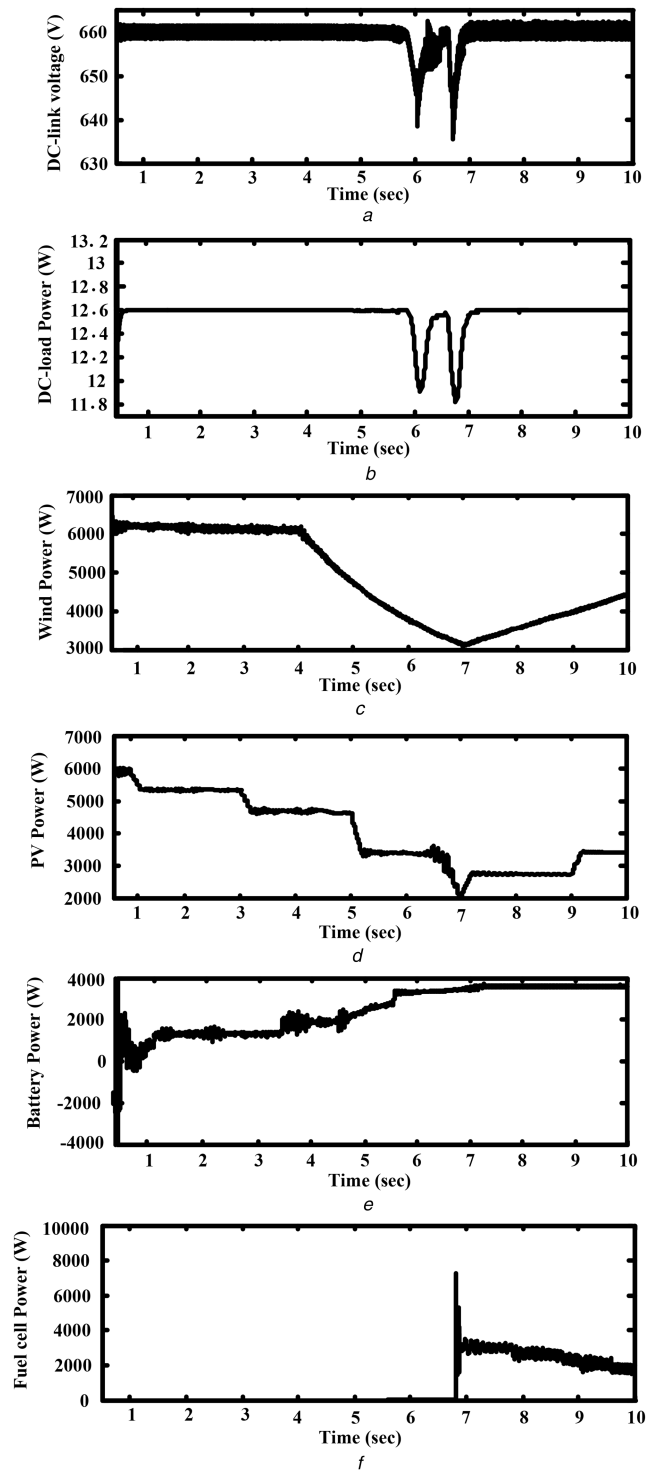


Fig. 14 Performance of priority-based load shedding algorithm during standalone mode of operation

(a) DC-link voltage, (b) Load power, (c) Wind power, (d) PV power, (e) Battery power, (f) FC power

other DC MG systems for different combinations of load and generation.

5 Conclusion

In this paper, an improved power management control strategy is proposed for improving both the dynamic and steady-state performance of the DC-link voltage in the DC MG. The combination of both voltage and SoC-based control strategy of the battery, FC, and electrolyser contributes a better power coordination between each source and load. The PI-controller gains of the BES system are tuned adaptively by using the TS-fuzzy control to get a better dynamic voltage response of the DC MG

system. The proposed proportional gain control technique of inverter-side controller regulates the modulation index and maintains the DC voltage within its prescribed limits during different power system scenarios such as the change in wind speed, variation in load, islanding mode, etc. The proposed priority-based load shedding algorithm maintains a proper power balance among different energy sources and storage devices during standalone or prolonged islanding mode of operation. The simulation results of the proposed control scheme are compared with the existing scheme and the superiority of the proposed power coordination management scheme of DC MG system is established.

The future research work may be about to implement different optimisation strategies in a cost-effective way during different system scenarios for energy management and increases the life of the battery in renewable energy-based DC MG. Also, the performance of the voltage stability and coordinated power control of DC MG with other energy storages such as supercapacitor, superconducting magnetic energy storage, in a large-scale renewable energy system, can be investigated.

6 References

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